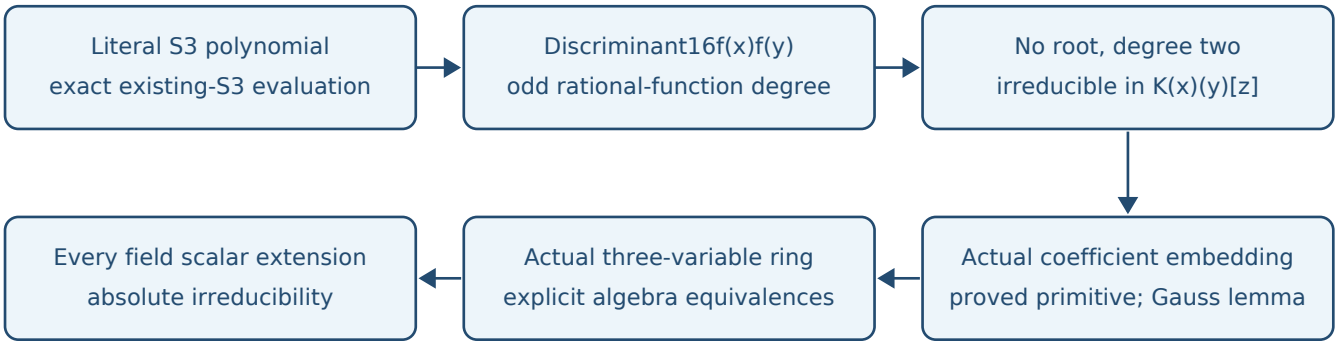
S3: symmetry and absolute irreducibility

Item1 | 13 November 2026 | Whole gate 64/0 failures/0 trusted | [pcc:unverified]

For every field K with2≠0 and arbitrary coefficients a,b, Lean proves the literal short-Weierstrass S3 polynomial is irreducible after every field homomorphism K→L. This includes algebraic closure. It also proves actual permutation symmetry and exact evaluation as the existing SemaevS3.S3 definition.



Exact source and symmetry. $S3=(x-y)^2z^2-2((x+y)(xy+a)+2b)z+(xy-a)^2-4b(x+y)$. In the elementary symmetric polynomials it equals $e2^2-4e1e3-2ae2-4be1+a^2$. The exact identity and Mathlib rename_esymm prove invariance under every variable permutation.

Function-field argument. Put $f(T)=T^3+aT+b$. The exact quadratic discriminant is$16f(x)f(y)$. In the actual nested rational-function field $K(x)(y)$, $f(x)$ is a nonzero coefficient and the y-degree of this discriminant is three. Every nonzero rational-function square has even integer degree, including rational functions with poles. Thus the quadratic has no root; its degree is two, so it is irreducible.

Actual polynomial bridge. The coefficient map $K[x,y]\to K(x)(y)$ is proved injective. The prime diagonal $y-x$ does not divide the middle coefficient: evaluating $y=x$ gives the nonzero polynomial $-4f(x)$. Every common divisor of the leading diagonal-square and middle coefficient is a unit, proving primitiveness. Gauss's injective-map theorem transfers irreducibility to $K[x][y][z]$. Explicit algebra equivalences then reach the actual MvPolynomial(Fin3) expression. The proof applies again after any field scalar extension.

Characteristic2 control. For the literal short formula over F2 at $a=b=0$, Lean proves $S3=((x+y)z+xy)^2$ and proves its factor is not a unit using evaluation at the origin. Hence this input is directly not irreducible. This is a restriction control, not a refutation of the qualified source theorem or the distinct ordinary characteristic2 elliptic model.

Proof map. Namespace SemaevIrreducibilityAudit: OddDegree.lean proves odd_degree_not_square; S3FunctionField.lean proves the actual discriminant/no-root/degree2 result; PrimitiveBridge.lean proves the coefficient embedding, primitive condition and nested irreducibility; ActualPolynomial.lean proves the actual MV/scalar-extension/symmetry/control certificate actual_S3_absolute_irreducibility_with_control.

Scope and assurance. This closes the n=3 algebraic symmetry/absolute-irreducibility gap. Higher arity, arbitrary-split normalization and other original14 obligations remain open. The source's short elliptic model assumes characteristic other than2/3 and nonsingularity; this algebraic statement needs only$2\neq0$. Lean4.35.0 / Mathlib56a005855ea17e0b0958f73ec801ae140b0a1679; standard propext, Classical.choice, Quot.sound only. No holes, custom axioms, native_decide, raised limits or waived checks. Local checks and independently signed standing are distinct. No new native experiment, paid compute or agent storage upload.

Primary source. Semaev, Summation polynomials and the discrete logarithm problem on elliptic curves, author manuscript5February2004, Theorem1 and page3; PDF SHA256 prefix1cab0ae67599, full URL/hash/read scope in evidence/source-review.json. The page3 overbar covers the whole function field K(X3), as visually checked; text extraction loses it. KB unavailable (MCP absent/wrapper127); bounded primary/repository/Mathlib fallback. No novelty claim. [pcc:unverified]